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The Alashan Terrane did not amalgamate with North China block by the Late Permian: Evidence from Carboniferous and Permian paleomagnetic results

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ABSTRACT

Rock magnetic and paleomagnetic studies have been carried out on the early Carboniferous limestones and the Late Permian purple sandstones sampled in the eastern Alashan Terrane (ALT), northwest of China. Two components were isolated from the Early Carboniferous limestone by thermal progressive demagnetisation: a low unblocking temperature component (LTC) of recent origin; a pre-folding medium temperature component (MTC) (the paleomagnetic pole is $\lambda = 13.1°$N, $\Phi = 11.0°$E, $A_{95} = 7.0°$) that is probably the result of the hydrothermal fluids from the Qilian Orgenic Belt acquired during the Late Carboniferous–Early Permian. Also, two components were separated from the Late Permian purple sandstone by thermal progressive demagnetisation: the LTC with the recent viscous remanent magnetisation, and the higher temperature component (HTC) revealed from three sections which has passed a regional fold test at the 95% probability level and reversal test, suggesting a primary characteristic magnetisation. The corresponding paleomagnetic pole is $\lambda = 27.2°$ N, $\Phi = 18.8°$ E, $A_{95} = 12.0°$. The apparent polar wander path (including early Carboniferous, late Carboniferous-Early Permian, Late Permian and Early–middle Triassic poles) of the ALT is significantly different with those of the NCB. Comparison of the APWPs between the ALT and NCB shows a strong similarity. If the APWP of Hexi Corridor–Alashan rotated counterclockwise around an Euler pole at 44°N, 84°E by 32°, then the coeval APW path of the ALT overlaps to that of the NCB. This result indicates that the ALT migrated to the NCB after the Early–Middle Triassic along a tectonic boundary located between Helanshan Mountain and Zhuozishan Mountain, and finally amalgamated to the NCB before the Early Cretaceous.

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1. Introduction

The continental crust of China is a mosaic of cratonic blocks and orogenic belts, and contains numerous small cratons and terranes that formed in various tectonic settings and have diverse origins, all of which began assembling in the Permian with amalgamation continuing through to the Cretaceous (Lin et al., 1985; Enkin et al., 1992; Yang and Besse, 2001; Zheng et al., 2013). A series of detailed paleomagnetic studies on the Tarim, North China (NCB), and Yangtze blocks, which dominate the tectonic framework of China, have been carried out over the last three decades (Zhao et al., 1990, 1992, 1993, 1996; Enkin et al., 1992; Huang et al., 1992; Yang et al., 1992, 1996, 1998, 2002, 2004; Yang and Besse, 2001; Zhu et al., 1998; Huang et al., 1999, 2000, 2001, 2005; Liu et al., 2010), leading to the establishment of Phanerozoic apparent polar wander paths (APWP). However, the very different pre-Cambrian basement and the different geological processes that operated during the Neoproterozoic meant that the existence of the so-called Sino–Korean–Tarim Block (Li et al., 1982; Ge et al., 2009) was not supported by this geological and paleomagnetic research. This is exemplified by the fact that the NCB paleomagnetic poles are far from the Tarim poles during the Carboniferous and Permian, and the APWP of the two blocks was near identical until the Late Jurassic (Zhu et al., 1998; Yang and Besse, 2001; Gilder et al., 2008), implying that amalgamation of the NCB and Tarim blocks must have occurred by this time. These terranes are largely covered by the Badain Jaran and Tengger deserts, meaning that the relationship between the

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Alashan (also called Alxa) Terrane (ALT), which bridges the Mongolian fold belt, and the Tarim, Qaidam, and North China blocks remains unclear. Some researchers consider the ALT to be the westernmost part of the NCB (e.g. Zhao et al., 2004, 2005; Santosh et al., 2007), and that collision of the Yinshan Block to the north of the ALT with the Ordos Basin formed the Western Block of the North China Craton (Zhao et al., 2004). However, more recent geological research suggests that the ALT did not form part of the NCB, primarily for the following reasons: (1) the results of *in situ* zircon U–Pb age data and Hf–O isotope analyses of samples from the Bayanwulashan and Diebusige complexes, two of the oldest rock units in the eastern Alashan Terrane, suggest that Archean rocks were not exposed in the eastern ALT (Dan et al., 2012); (2) geochronological analyses of detrital zircons from Ordovician and Devonian sediments in the eastern ALT yielded very different age populations and corresponding Hf isotope data compared with similar units of similar ages in the NCB, suggesting that the ALT was not connected with the NCB prior to the Middle Devonian (Zhang et al., 2011; Yuan and Yang, 2014); (3) field geological observations favour amalgamation of the ALT with the NCB between the Late Ordovician and the Early Devonian (Li et al., 2012; Zhang et al., 2013) or during the Late Hercynian–Indosinian (Late Triassic; based on a tectono-stratigraphic comparison of the ALT and the NCB; Ge et al., 2009); and (4) the Late Devonian amalgamation of the ALT, the Qilian Orogenic Belt and the Qaidam Block implies a close relationship between the amalgamated ALT–Qaidam and Tarim blocks during the late early Paleozoic (Xiao et al., 2009a).

These contrasting models indicate that the ALT is a key region in terms of understanding the tectonic evolution of Northern China and Central Asia (Fig. 1). Understanding the late Paleozoic tectonic relationship between the ALT, NCB, and Tarim blocks will further our knowledge of the tectonic evolution of Northern China, enabling the identification of key regional geological processes. Here, we present the results of a paleomagnetic study of early Carboniferous and late Permian units in the eastern ALT and use these data to constrain the timing of amalgamation of the ALT and the NCB.

2. Geological setting and paleomagnetic sampling

The ALT is separated from the Ordos Basin of the NCB to the east by either the western marginal fault of the Ordos Basin (or the Zhuozishan Fault) or the Helan Shan Fault, from the Tarim Block to the west by the Altyn Taugh Fault, from the North Qilian Fold Belt to the south by the north boundary fault of the belt, and is connected with the Central Asian Fold Belt to the north by the Engeerwusu ophiolitic melange belt (Fig. 1; Huo et al., 1989; Regional Geology of Ningxia Hui Autonomous Region, 1990; Wang et al., 1992; Wu and He, 1992; Zhao et al., 1993; Li et al., 1999; Huang et al., 2000; Lu et al., 2009). The ALT is dominated by late Archean and Paleoproterozoic metamorphic basement units; zircon U–Pb dating of these units indicates that the lower part of the Diebusige Complex formed at ca. 2.7 Ga and the Bayanwulashan Complex formed at ca. 2.5–2.26 Ga (Geng et al., 2007; Dan et al., 2012). Syn-collisional granites were intruded at 971 and 845 Ma in the western part of the ALT (Geng et al., 2002), suggesting that this area was affected by a Neoproterozoic thermo-tectonic event during the assembly and break-up of Rodinia. These Archean rocks are covered by the Lower to Middle Cambrian Haobiru Formation, a series of siliceous rocks intercalated with phosphatic horizons, and a continuous Middle–Upper Cambrian to Silurian sedimentary succession that was deposited on the northern part of the ALT, and

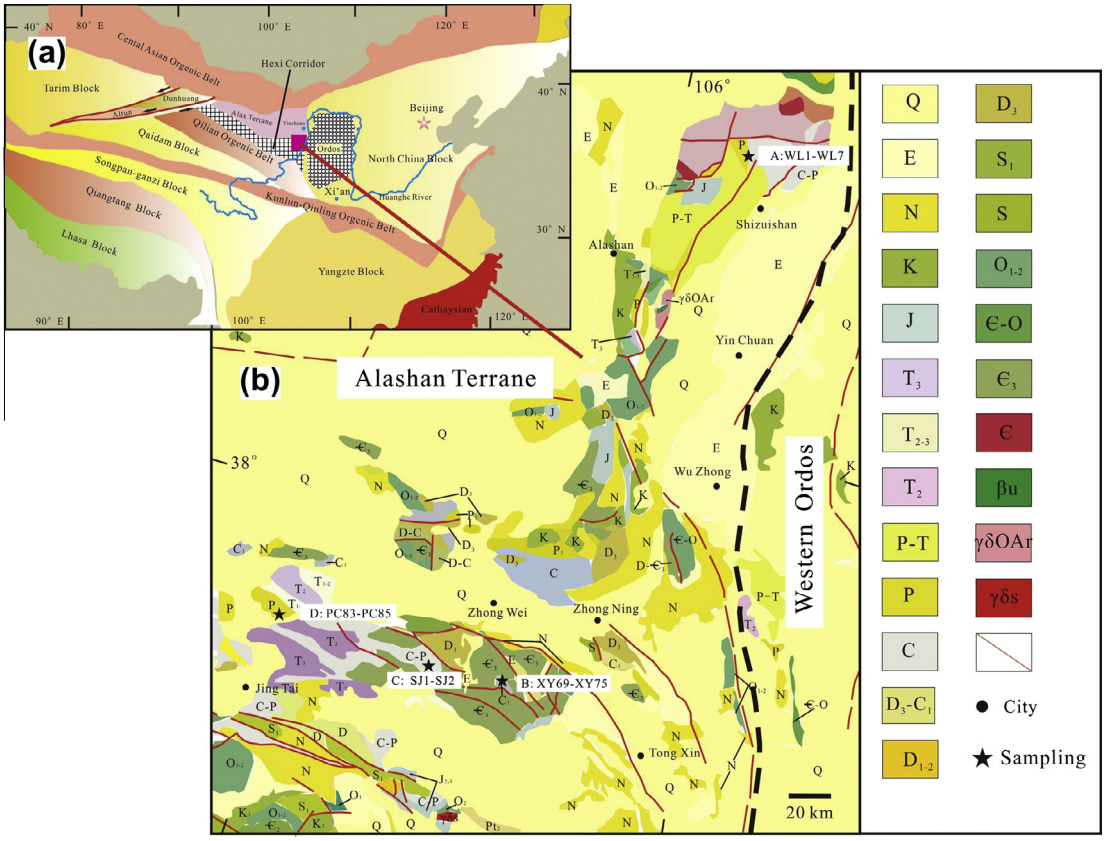


Fig. 1. (a) Tectonic subdivisions of the main continental blocks and orogenic belts of eastern Asia. The sampling area in the southeastern part of Gansu Province and within the Ningxia Hui Autonomous Region is indicated by a red square. (b) Geological sketch map of the sampling areas; locations of paleomagnetic sampling sections are shown as stars. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

consists of shallow–marine shelf facies carbonate to hemipelagic sediments (Zheng and Zhu, 1987).

Carboniferous sediments in the study area are divided into, in ascending order, the lower Carboniferous Chouniugou and Qianheishan formations, the middle Carboniferous Jingyuan and Yanghugou formations, and the upper Carboniferous Taiyuan Formation in the southeastern ALT. The Chouniugou Formation is divided into two parts based on lithological and lithofacies characteristics: a lower section that is dominated by grey quartz sandstones intercalated with grey shales, carbonate shales, and siltstones, and that includes a basal section containing coarse-grained quartz sandstones intercalated with conglomerates, and an upper section containing locally outcropping thin-bedded marine marls. The upper section of the formation contains grey to dark grey marine shales intercalated with marls, thin-bedded bioclastic limestones, carbonaceous shales, and siltstones. Carbonates are commonly found in this formation in the areas around Heishan and Xiaoweichuan. The Chouniugou Formation is disconformably underlain by the early Carboniferous Qianheishan Formation and is conformably or disconformably overlain by the middle Carboniferous Yanghugou Formation. The lower part of the Chouniugou Formation contains a uniform plant-based biota, which contains few brachiopods, indicating deposition in an offshore lake–swamp environment. The shales and limestones in the upper part of the formation are rich in marine fossils, including corals, brachiopods, bivalves, gastropods, ammonoids, bryozoans, and crinoids, indicative of deposition in a shallow-marine littoral sedimentary environment in a period with a warm climate (Huo et al., 1989). The Chouniugou Formation has been dated using these corals and brachiopods, yielding a middle–late Visean age for the lower part of the formation, an early Namurian age for the central part, and a middle–late Namurian age for the upper part (Shen et al., 1994). The early Carboniferous samples analysed during this study were collected from the Chouniugou Formation in the Zhongwei area (37.2°N, 105.4°E) of the Ningxia Hui Autonomous Region, to the southeast of the ALT (Fig. 1b). A total of 80 samples from 7 sites (XY69–XY75) were collected from grey–dark grey, thin to moderately bedded limestones in the upper section of the Chouniugou Formation using a portable drill and with samples oriented using a magnetic compass. Four sites (XY69 to XY72) were sampled along one flank of a fold, with the other three sites (XY73 to XY75) sampled on the other flank.

Upper Permian sediments in the Helanshan area are allocated to the Shiqianfeng Formation (P2sh), a series of thinly bedded grey–light grey to green–purple, fine- to coarse-grained feldspar quartz sandstones, purple siltstones, sandy shales, and mudstones, with an upper part that is interbedded with rhyolitic and dacitic tuffs and a lower part that includes marl lenses or grey calcareous nodules. The pollen assemblage in this formation includes Florinites sp., Cordaitina sp., Verrucosisporites sp., Trachytrildes sp., and Lycopodiumporites, indicating an upper Permian age of deposition. The Shiqanfeng Formation is conformably underlain by the upper Permian Shihezi Formation, and is conformably overlain by Triassic sediments (Regional Geology of Ningxia Hui Autonomous Region, 1990). A total of 64 purple siltstone samples were obtained from seven sites in the Shiqianfeng Formation within the Wulankuang section of the northeastern Helanshan area (39.2°N, 106.3°E; Fig. 1b). Upper Permian sediments in the Hexi Corridor area are assigned to the Yaogou Group (P2yg), which is dominated by purple, miscellaneous greyish–green, and grey feldspar and quartz sandstones, quartz sandstones, siltstones, mudstones, tuffaceous sandstones and minor tuff layers that are indicative of the accumulation of lacustrine and terrestrial volcanic debris. The Yaogou Group is conformably overlain by the lower Permian Dahuanggou Group (Huo et al., 1989). A total of 22 fine-grained purple feldspar quartz sandstone samples were collected from two sites in the Yaogou Group within the Shenjing–Sanyanjing section (37.1°N, 105.0°E), with a further 27 fine-grained purple sandstone samples collected from three sites in the Baidunzi section, in the Jingtai area (37.3°N, 104.1°E) of Gansu Province (Fig. 1b). All samples were collected with a portable gasoline-powered drill and were oriented using a magnetic compass.

3. Paleomagnetic experiments and results

The laboratory analysis undertaken during this study involved the progressive thermal demagnetisation of all samples over 16–20 steps using an ASC TD-48 thermal demagnetiser. Natural remanent magnetisation (NRM) was measured using a 2G-755 cryogenic magnetometer installed inside a shielded room at the paleomagnetic laboratory of Nanjing University, China, and the anisotropy of magnetic susceptibility (AMS) was measured using an AGICO KLY-3s Kappabridge. Isothermal remanent magnetisation (IRM) was used to identify magnetic minerals within the samples by applying three different magnetic fields along three orthogonal axes (2.40, 0.40, and 0.15 T along the *Z*, *Y*, and *X* axes, respectively) before thermal demagnetising the IRMs (Lowrie, 1990). Temperature-dependent magnetic susceptibility experiments were conducted using a KLY3s Kappabridge and a CS-3 high-temperature furnace to test magnetic mineral transformations during laboratory heating. The demagnetisation result of each sample was analysed using principal component analysis (Kirschvink, 1980), and site-mean directions were calculated by Fisherian statistics (Fisher, 1953). All of the data were analysed with the paleomagnetic software developed by Enkin (1994) and Cogné (2003).

3.1. Early Carboniferous

Initial rock magnetic experiments were undertaken to identify magnetic minerals within representative limestone samples. Temperature dependences of magnetic susceptibility (k) experiments obtained for these samples show some evidence for a Hopkinson peak near 580 °C on the heating curves, indicative of magnetite (Dunlop and Özdemir, 2001; Fig. 2a and b). The magnetic susceptibility values for the cooling curves are higher than the values for the heating curves at temperatures higher than 580 °C, a feature that is indicative of changes in strong magnetic minerals during heating. IRM experiments identified a soft coercivity magnetic mineral at 60–100 mT, potentially indicative of titanomagnetite (Fig. 2f and g), and a coercivity around 200–300 mT that may be indicative of composite magnetite and hematite (Fig. 2g). Thermal demagnetisation of composite IRMs indicates the presence of hematite (Fig. 2e). Magnetic hysteresis curves were measured using a MicroMag 2900 instrument at the Department of Physics, Peking University, China; this analysis yielded magnetic hysteresis curves with wasp-waisted behaviour (Fig. 3), suggesting a possible mixture of different coercivity minerals. Saturation magnetisation of ferromagnetic components (Ms), saturation remanent magnetisation (Mrs), and coercivity (Bc) values were determined after correcting for high field paramagnetic contributions. The majority of samples plot in the pseudo-single-domain (PSD) field of a Mrs/Ms versus Bcr/Bc ratio Day diagram (Day et al., 1977; Fig. 4). These data indicate that magnetic carriers within the limestone are hematite, magnetite, and/or titanomagnetite with PSD grain sizes.

Thermal demagnetisation was conducted over 12–17 progressive steps, with temperature intervals of 50 °C below 300 °C, 30–40 °C between 340 °C and 420 °C, and 10–20 °C above 420 °C. Two components were isolated from all samples obtained from the seven sites sampled during this study (Fig. 5a–e). A lower temperature component (LTC) was isolated at temperatures lower

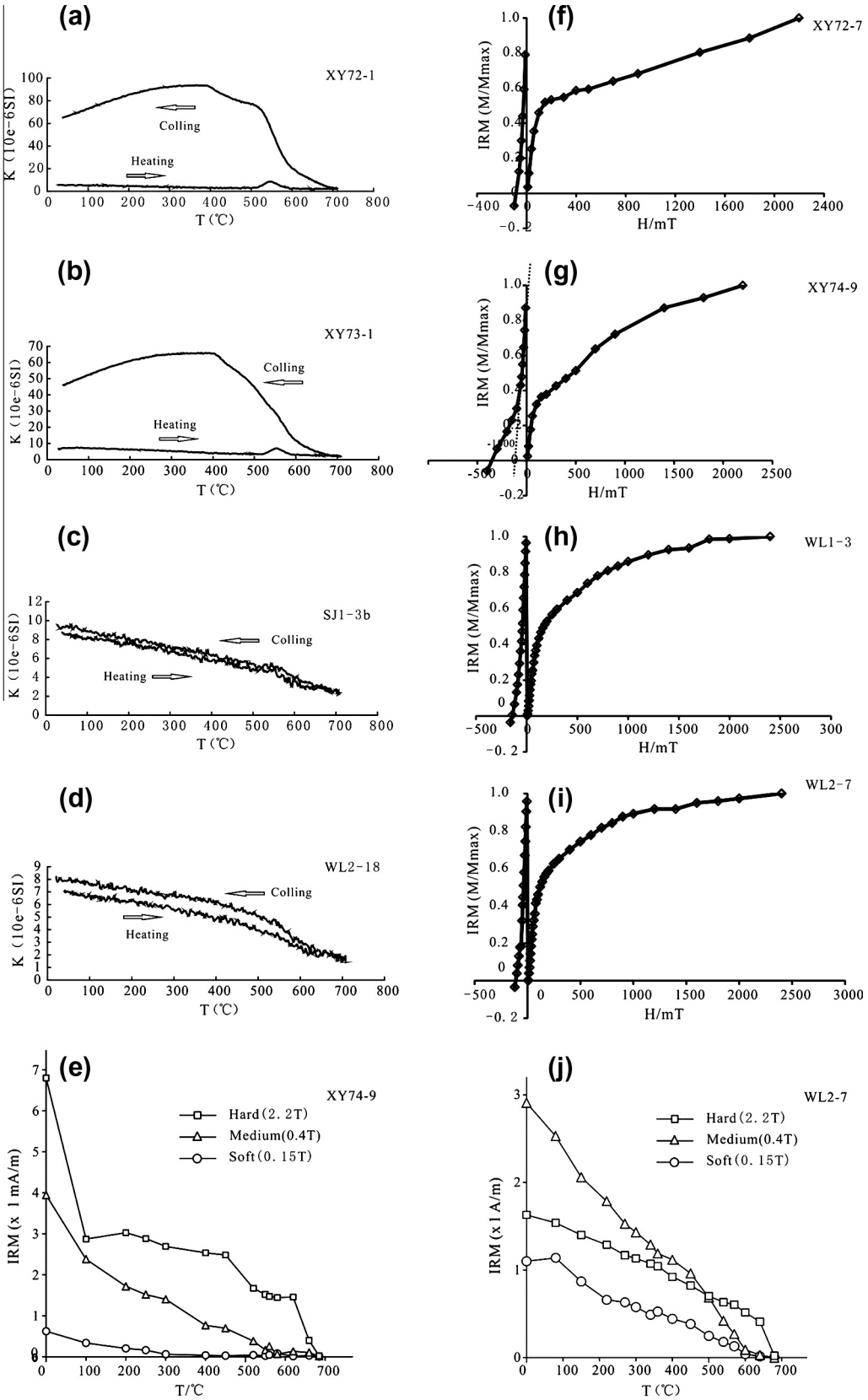


Fig. 2. (a–d) Thermomagnetic curves of magnetic susceptibility of representative samples. (f–i) Behaviour of acquisition and back-demagnetisation of IRM for representative samples. (e and j) Progressive thermal demagnetisation curves of composite IRMs for representative samples.

than 240 °C, yielding a site-mean direction of Dg = 7.0°, Ig = 60.8°, kg = 452.2, and α_{95} = 2.8° before bedding correction, and Ds = 264.9°, Is = 68.4°, ks = 3.6, and α_{95} = 36.8° after bedding correction (Fig. 6a and Table 1). The LTC produces a negative fold test at a 99% significance level (McFadden, 1990), indicative of remanent acquisition after folding and possibly the viscous

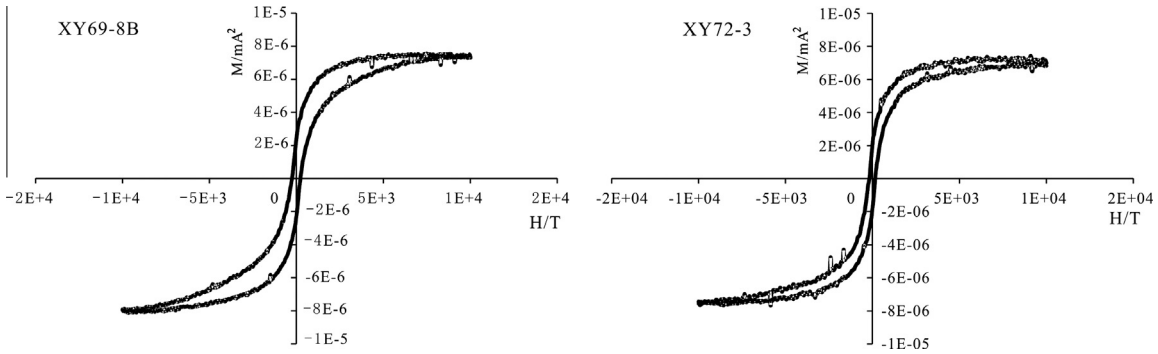


Fig. 3. Magnetic hysteresis loop of early Carboniferous Chouniugou Formation limestone samples; H = DC field intensity, M = remanent magnetisation.

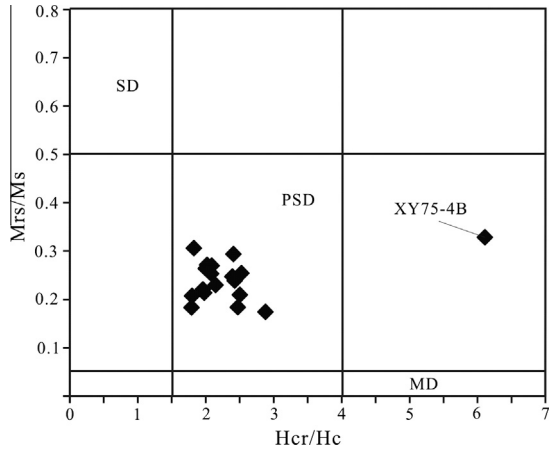


Fig. 4. Day diagram for early Carboniferous Chouniugou Formation limestone samples (Day et al., 1977); SD = single domain; MD = multi-domain, PSD = pseudo-single-domain.

remanent magnetisation of the present-day geomagnetic field. Another linear medium temperature component (MTC) was separated between 300 °C and 460 °C, yielding a site-mean direction at Dg = 116.9°, Ig = −18.9°, kg = 18.8, and α_{95} = 14.3° before bedding correction, and Ds = 103.0°, Is = −9.0°, ks = 47.0, and α_{95} = 8.9° after bedding correction (Fig. 6b; Table 2). The MTC passes McFadden's fold test at a 95% confidence level (McFadden, 1990; critical value ξ_c = 3.086, ξ_1 = 4.900 in geographic coordinates, and ξ_2 = 7.318 in stratigraphic coordinates), indicating that the MTC direction was obtained prior to the Middle Jurassic folding event. The results of thermal demagnetisation indicate that the MTC was unblocked near 460 °C, and did not tend to the origin (Fig. 5a–e). The results of AF demagnetisation of sample XY70-7 also show that demagnetised directions change along a great circle after the lower coercivity (or temperature) component is removed (Fig. 5b and c), with stable end-point directions absent, indicating that a HTC component as high as 90 mT could be hidden. This suggests that an HTC may be present that cannot be isolated after 460 °C, either because magnetic minerals were transformed, yielding erratic remanent directions, or as a result of very weak remanent magnetisation. The result suggests that the MTC is a secondary remanence that was acquired before the Middle Jurassic. A mean paleomagnetic pole calculated from the MTC is at λ = 13.1°N, Φ = 11.0°E, A_{95} = 7.0°, with a paleolatitude of 4.5°N ± 7.0°.

3.2. Late Permian

The *k–T* curves for late Permian samples analysed during this study are indicative of the presence of hematite (Fig. 2c and d). The IRM measurements undertaken during this study increase rapidly with increasing external direct field values under 100 mT, but increase slowly at values >100 mT, and do not reach saturation until 2400 mT (Fig. 2h and i). The coercivity of remanence deduced from back-demagnetisation below 100 mT is indicative of the predominance of a low-coercivity magnetic mineral such as titanomagnetite. The results of three-axis thermal demagnetisation of IRMs show that both medium and soft components are unblocked at 580 °C, indicative of magnetite, and that hard, medium, and soft components are all unblocked at 680 °C, indicative of hematite (Fig. 2j). This suggests that the dominant magnetic carrier within the late Permian purple sandstone is hematite, with minor magnetite and/or probably titanomagnetite.

Thermal demagnetisation was conducted over 14–16 progressive steps, with temperature intervals of 80–100 °C below 500 °C, 20–30 °C between 500 °C and 600 °C, and 10–20 °C at temperatures higher than 600 °C (Fig. 5f–o). Two components were isolated for all of the samples obtained from the three sections sampled during this study (Fig. 5f–o). The site-mean direction of the LTC for the 12 sampling sites is Dg = 14.0°, Ig = 66.6°, kg = 148.2, and α_{95} = 3.4° before bedding correction, and Ds = 19.9°, Is = 45.5°, ks = 2.1, and α_{95} = 38.7° after bedding correction (Fig. 7a; Table 3). The LTC yields a negative fold test at a 99% significance level (McFadden, 1990), indicating a post-folding origin and possibly the viscous remanent magnetisation of the present-day geomagnetic field. The HTC isolated between 580 °C and 680 °C for the remaining nine sites are consistent. The site-mean direction is Dg = 293.7°, Ig = 25.8°, kg = 3.0, α_{95} = 35.9° before bedding correction, and Ds = 289.7°, Is = 33.8°, ks = 21.5, α_{95} = 11.4° after bedding correction (Fig. 7b; Table 4). The fold test is positive at a 99% confidence level for the data from the Wulankuang section (with values of ξ_1 = 0.64 in geographic coordinates, ξ_2 = 0.28 in stratigraphic coordinates, and a critical value ξ_c = 0.36), and a positive fold test at a 95% confidence level was obtained for the data from all three sections (with ξ_1 = 0.72 in geographic coordinates, ξ_2 = 0.28 in stratigraphic coordinates, and a critical value ξ_c = 0.35). These results indicate that the magnetic remanence of these samples was acquired prior to the Middle Jurassic folding event and contrasts sharply with the paleomagnetic results for Triassic and younger units in the study area (Liu et al., 2010), indicative of a primary origin for this magnetism. The average Permian paleomagnetic pole for the nine sites within the ALT is at λ = 27.2°N, Φ = 18.8°E, and A_{95} = 12.0°, with a paleolatitude of 18.6°N ± 12.0°.

4. Discussion

4.1. Remagnetisation of Carboniferous limestone

Two previous studies have focused on the identification of a Carboniferous paleomagnetic pole in the Zhongwei area of the ALT (He et al., 1988; Huang et al., 2001). He et al. (1988) reported

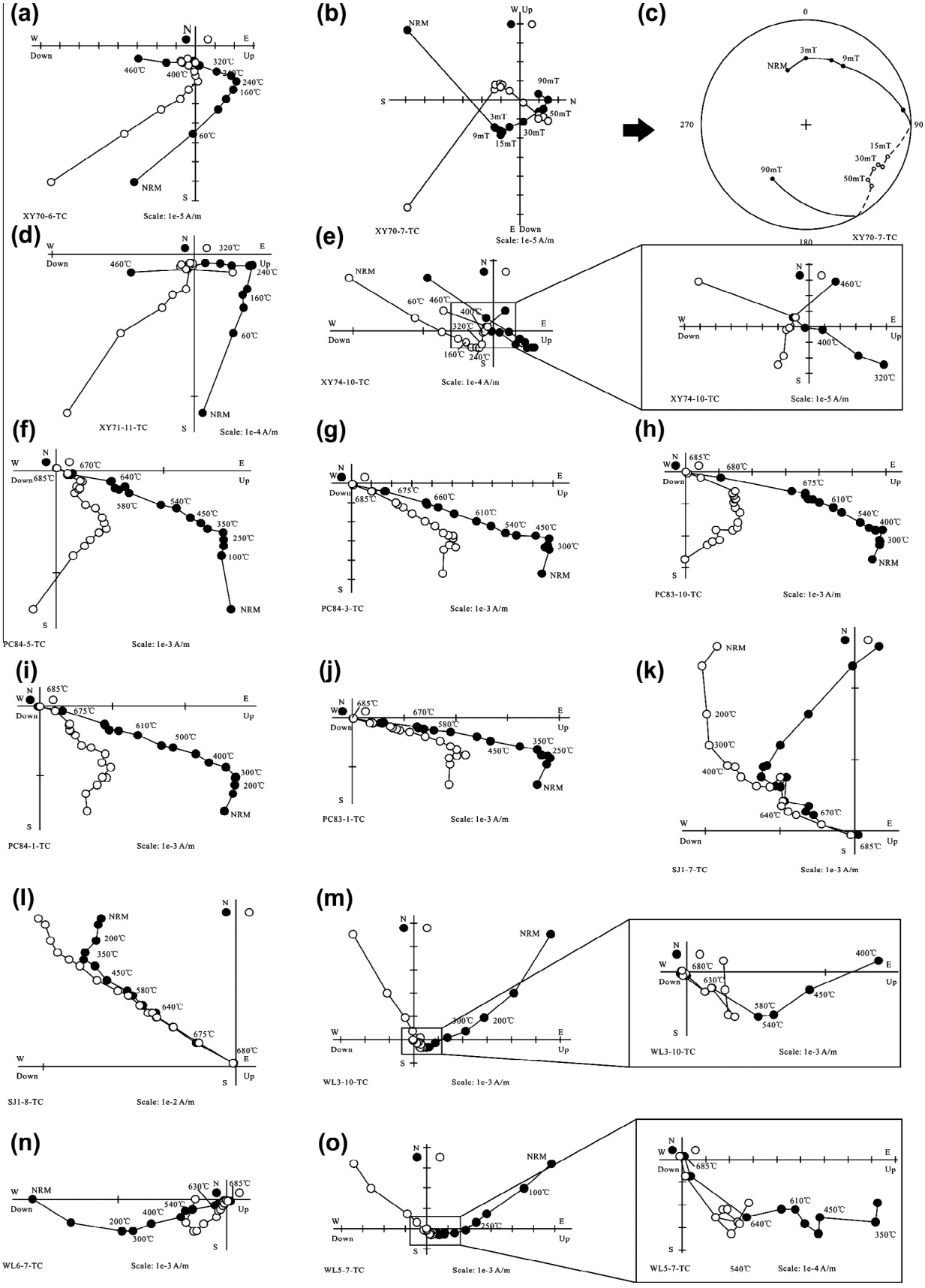


Fig. 5. Zijdeveld and stereo-plot diagrams for the results of thermal and/or alternating field demagnetisation of representative samples, with directions plotted after tilt corrections. Open and solid circles represent vector endpoints projected onto vertical and horizontal planes, respectively. (a–e) Early Carboniferous limestone samples; (f–o) late Permian purple sandstone samples.

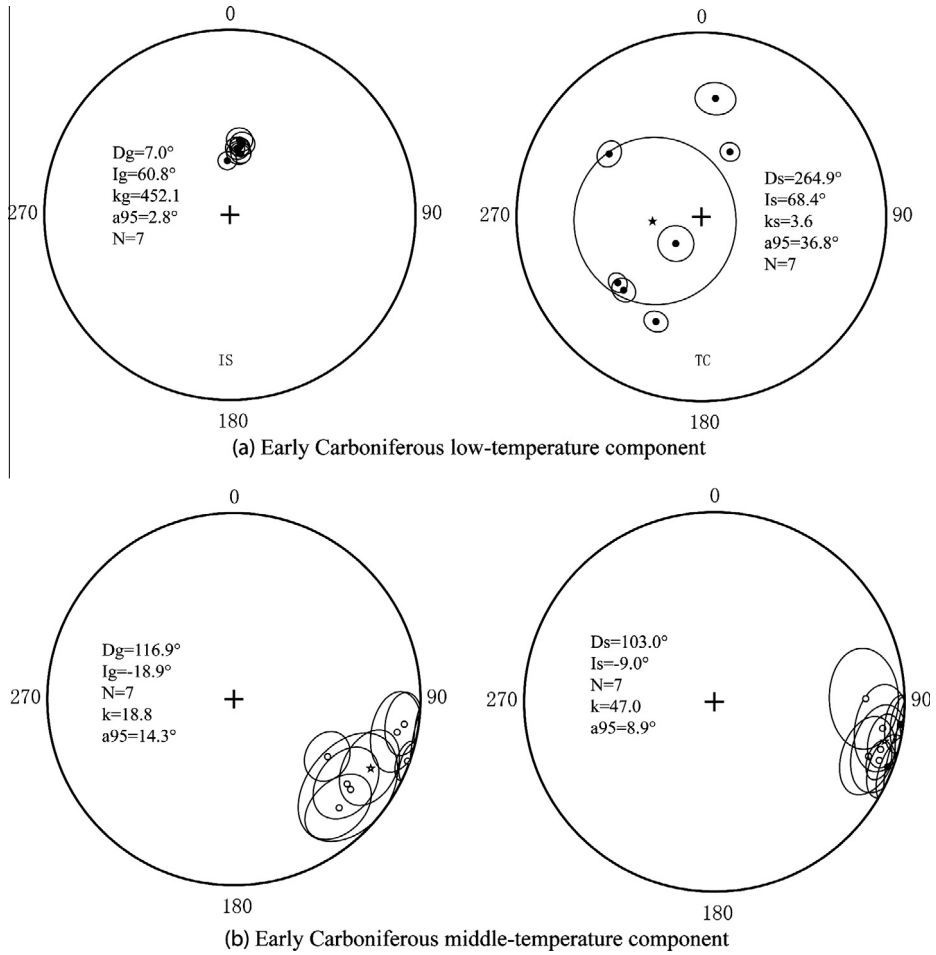


Fig. 6. Equal-area projections of (a) site-mean low-temperature component directions for the early Carboniferous and (b) site-mean middle-temperature component directions before and after tilt corrections; solid/open symbols represent downward/upward dipping directions, respectively, shown with 95% confidence limits. The star indicates the mean direction. IS = *in situ*, TC = tilt-corrected.

Table 1
Summary of sampling data and result for the low-temperature component from the Early Carboniferous limestone.

Site ID	Strike/Dip	R/N (N0)	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95}	Φ(G)	λ(G)	Φ(S)	λ(S)
Xiaoweichuan section: 37.2°N, 105.3°E												
XY69	126/72	0/8(10)	10.1	62.7	231.7	41.9	176.6	4.2	150.1	79.7	58.3	−11.7
XY70	122/73	0/7(10)	8.1	62.3	226.6	41.7	134.6	5.2	146.4	81.1	62	−14.7
XY71	104/82	0/15(17)	6.3	60.3	–	–	58.4	5.0	153.7	83.7	–	–
			–	–	203.4	38.1	61.4	4.9	–	–	80.7	−27.4
XY72	064/9	0/8(9)	10	57.9	–	–	112.4	5.2	182.5	82	–	–
			–	–	23.7	58.4	175	4.2	–	–	182.2	71.3
XY73	250/39	0/9(9)	8.1	59.6	–	–	141.4	4.3	165.6	82.9		
			–	–	6.8	35.5	42.1	8.0			264.8	71.5
XY74	171/52	0/9(10)	7.3	56.4	304.4	39.4	88.8	5.5	195.5	84.2	14.3	40.2
XY75	112/24	0/12(12)	357.3	66.1	–	–	109.6	4.0	96.2	78.6		
			–	–	223.7	73.7	28.6	7.9			84.3	13.4
Average		7	7.0	60.8	–	–	452.2	2.8	152.6	82.9		
			–	–	264.9	68.4	3.6	36.8			62.1	25.5

Abbreviations: Site ID: site identification; Strike/dip: strike azimuth and dip of bed; R/N: samples show reversed and normal polarity; N0: number of samples in the calculation; Dg, Ig (Ds, Is): declination and inclination of direction *in situ* (after tilt-correction); Φ(G), λ(G) (Φ(S), λ(S)), longitude and latitude of VGP in geographical (stratigraphic) coordinates.

a Carboniferous paleomagnetic pole (27.2°S, 163.7°E) for the Xiangshan area, although this was only a preliminary result, primarily as only 15 samples from three sites were collected during this study, yielding results that were questioned by the authors themselves. In comparison, the careful rock magnetic and demagnetisation analysis undertaken by Huang et al. (2001) enabled the isolation of a higher-temperature component between 350 °C and ~530–550 °C from Carboniferous limestones within the Xiangshan area. The characteristic components of these limestones passed the fold test, and yielded an early–middle Carboniferous pole at λ = 10.5°N, Φ = 14.0°E, and A_{95} = 6.2° (N = 10 sites, 39 samples), and a preliminary late Carboniferous pole at λ = 33.3°N, Φ = 10.2°E, and A_{95} = 16.7° (N = 4 sites, 16 samples). Although the paleomagnetic pole calculated from our MTC results is compatible

Table 2
Summary of sampling data and results for the Middle temperature component from the Early Carboniferous limestone.

Site ID	Strike/Dip	R/N (N0)	Dg (°)	Ig (°)	Ds (°)	Is (°)	*k*	α_{95}	Φ(G)	λ(G)	Φ(S)	λ(S)
Lower Carboniferous Chouniugou Formation, Xiaoweichuan section: 37.2°N, 105.3°E												
XY69	126/72	7/0(10)	136.0	−19.9	109.5	−15.1	21.6	13.8	172.6	−42.2	189.6	−20.1
XY70	122/73	5/0(10)	126.9	−25.1	99.0	−11.4	29.9	14.6	182.9	−37.1	194.5	−10.6
XY71	104/82	4/0(17)	127.8	−21.9	–	–	18.3	22	180.4	−36.7		
			–	–	88.8	−21.5	27.2	17.9			205	−5.8
XY72	064/9	7/0(9)	109.7	−1.3	–	–	87.7	6.5	183.6	−16		
			–	–	109.6	−8.8	84.2	6.6			186.9	−18.2
XY73	250/39	6/0(9)	101.5	−11.7	–	–	24.6	13.8	193.1	−12.7		
			–	–	106	−10	21.4	14.8			189.7	−15.8
XY74	171/52	7/0(10)	121.6	−40.7	110.6	3	35.3	10.3	285.3	29.5	181.4	−15.3
XY75	112/24	12/0(12)	98.4	−8.5	–	–	18	10.5	193.6	−9.3		
			–	–	97	1.6	24	9			190.5	−5.1
Average		*N* = 7	116.9	−18.9			18.8	14.3	186.6	−27.2		
					103.0	−9.0	47.0	8.9			191.0	−13.1

Abbreviations: Site ID: site identification; Strike/dip: strike azimuth and dip of bed; R/N: samples show reversed and normal polarity; N0: number of samples in the calculation; Dg, Ig (Ds, Is): declination and inclination of direction *in situ* (after tilt-correction); Φ(G), λ(G) (Φ(S), λ(S)), longitude and latitude of VGP in geographical (stratigraphic) coordinates.

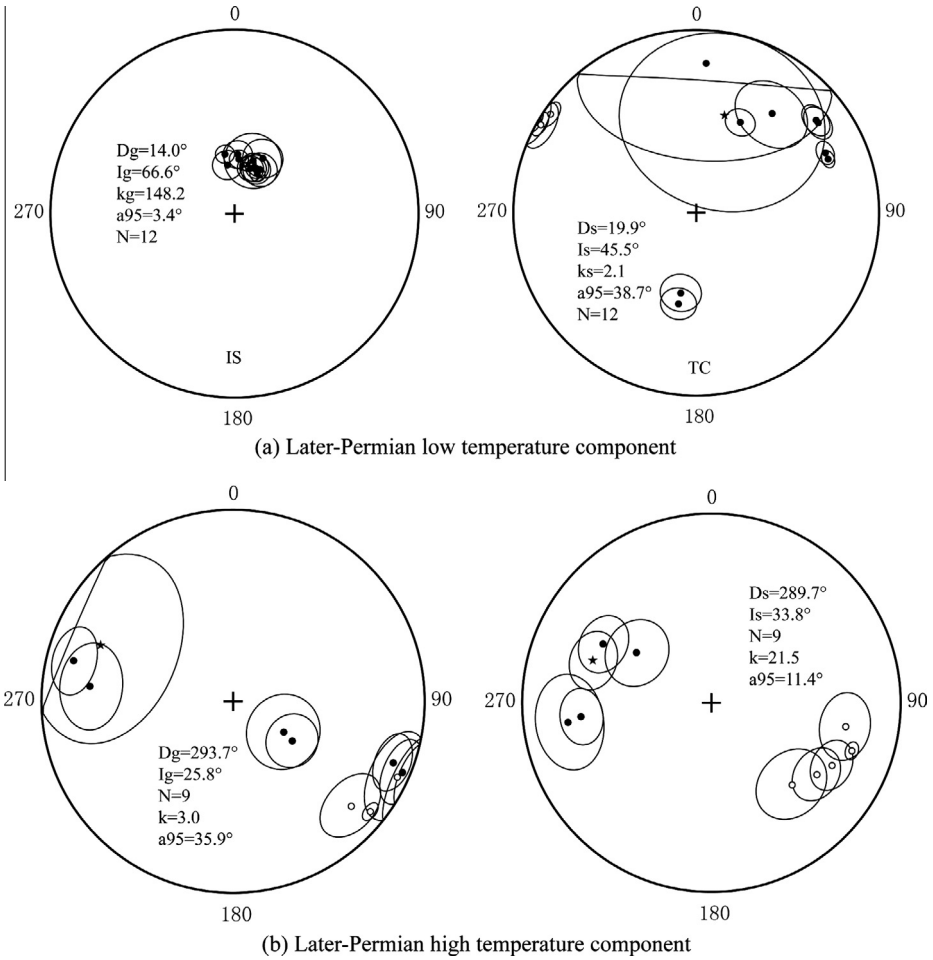


Fig. 7. Equal-area projections of (a) site-mean low-temperature component directions from late Permian samples, and (b) site-mean directions of the high-temperature component before and after tilt correction; symbols are as in Fig. 6.

with the early–middle Carboniferous pole of Huang et al. (2001), our pole contrasts sharply with their late Carboniferous pole and other post-late Permian poles in the study area. A positive fold test indicates that the remagnetised MTC was acquired prior to the Middle Jurassic, most probably during the late Carboniferous–early Permian.

Limestone remagnetisation could have occurred during interaction with orogenic fluids during continental collision (McCabe et al., 1984), a process that may also have destroyed the original magnetic fabric of these sediments (Addison et al., 1985). The sampling areas are located in the eastern part of the ALT, an area that recorded the pre-Late Devonian closure of the North Qilian Ocean and subsequent Carboniferous orogenesis and orogenic fluid activity. Recent mineral exploration in this area has led to the identification of epithermal gold and copper mineralisation in the lower part of the late Carboniferous Tupo Formation, which unconformably overlies the early

Table 3
Summary of sampling data and results for the low-temperature component of the Late Permian samples.

Site ID	Strike/Dip	R/N (N0)	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95}	Φ(G)	λ(G)	Φ(S)	λ(S)
Upper Permian Shiqianfeng Formation, Wulankuang section: 39.2°N, 106.3°E												
WL1	356/54	0/8(9)	25.5	67.8	65.1	23.2	155.5	4.5	154.9	68.7	202.6	26.8
WL2	356/54	0/8(10)	30.2	69.8	67.9	23.2	283	3.3	151.1	65	200.7	24.7
WL3	339/55	6/0(11)	23.1	67.7	52.3	18.3	59	6.3	153.5	70.2	214.4	34.8
WL4	339/55	0/7(11)	25.7	68.3	53.6	18.2	55.7	6.5	153.3	68.3	213.4	33.8
WL5	339/55	0/11(11)	350.9	63.3	–	–	126.6	4.1	59.1	81.2		
			–	–	304.1	-4.3	101.7	4.5			351.5	24.2
WL6#	184/88	0/8(8)	4.6	65.8	–	–	187.8	4.1	125.4	80.5		
			–	–	299.6	2	183.3	4.1			357.3	23.2
WL7#	188/86	0/8(8)	351.8	68.3	–	–	73.2	6.5	84	76.5		
			–	–	299.5	-2.7	47.3	8.1			355.6	21.5
Upper Permian Yaogou Group, Shengjing-Sanyanjing section: 37.1°N, 105.0°E												
SJ-1#	325/35	0/8(11)	3.2	63.7	26	44.2	78.9	6.3	120.3	81.4	214.4	65.3
SJ-2#	325/40	0/7(11)	19.3	65.1	–	–	24.6	12.4	153.8	72.6		
			–	–	37.4	32.2	16.6	15.3			215.6	51.8
Upper Permian Yaogou Group, Baiduizi section: 37.3°N, 104.1°E												
PC83	108/64	0/7(8)	30.3	67.5	191	47.9	66.4	7.5	153.2	64.8	93.7	-22.9
PC84	108/64	0/7(8)	27.3	62.4	190.9	53.1	48.9	8.7	168.2	68.4	94.6	-18.3
PC85	15/32	0/11(11)	359.7	59.1	49.5	53.1	143.9	3.8	99	87.4	183.9	50
	Average	N = 12	14.0	66.6	–	–	148.2	3.4	140.7	74.3		
			–	–	19.9	45.5	2.1	38.7			220.4	70.4

Abbreviations: Site ID: site identification; Strike/dip: strike azimuth and dip of bed (# denotes an average value); R/N: samples show reversed and normal polarity; N0: number of samples in the calculation; Dg, Ig (Ds, Is): declination and inclination of direction *in situ* (after tilt-correction); Φ(G), λ(G) (Φ(S), λ(S)), longitude and latitude of VGP in geographical (stratigraphic) coordinates.

Table 4
Summary of sampling data and results for the high temperature component (HTC) of the Late Permian samples.

Site ID	Strike/Dip	R/N (N0)	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95}	Φ(G)	λ(G)	Φ(S)	λ(S)
Shiqianfeng Formation, Wulankuang section: 39.2°N, 106.3°E												
WL1*	356/54	0/9(9)	353.8	42.3	31.5	24.7	4.1	31.1	307.7	74.4	230.9	51.8
WL2*	356/54	0/10(10)	0.2	54.6	45.9	26.4	3.2	34.1	284	85.9	215.3	42.5
WL3	339/55	6/0(11)	111	11.4	117.5	−28.9	28.8	9.8	178.2	−12.3	190.8	−30.5
WL4	339/55	10/0(11)	112.8	4.6	124.3	−33.0	22.7	10.7	179.6	−15.9	189.1	−37.6
WL5	339/55	6/0(11)	114.8	−6.2	135.5	−39	20.8	15	182.5	−21	185.2	−48.5
WL6	184/88	0/7(8)	121.3	64.7	–	–	15.5	15.8	143.0	10.5	–	–
			–	–	262.2	24.5	13	17.4			31.1	2.2
WL7	188/86	0/7(8)	123.8	59.5	–	–	29.8	11.2	145.8	4.6	–	–
			–	–	263.9	31.3	33.6	10.6			33.2	6
Shengjing-Sanyanjing section: 37.1°N, 105.0°E												
SJ-1	325/35	0/10(11)	284.3	15.2	–	–	18.1	11.7	12.6	16.0	–	–
			–	–	298.5	35.1	19.0	11.4			14.5	34.0
SJ-2	325/40	0/6(11)	276	25.8	–	–	19.2	15.7	22.4	12.9	–	–
			–	–	303.8	50.6	23.8	14			25.2	43.8
Upper Permian Yaogou Group, Baiduizi section: 37.3°N, 104.1°E												
PC83	108/64	8/0(8)	131.6	−18.6	100.1	−28.7	20.3	12.6	174.4	−38.4	200.7	−17.1
PC84	108/64	8/0(8)	128.8	−8.9	108.9	−22.6	276.8	3.3	171.9	−33	192.4	−22.1
PC85*	15/32	0/6(11)	358.2	53	40.6	50.3	30.2	12.9	306.1	86	191.8	56.1
Average		N = 9	293.7	25.8			3.0	35.9	18.1	29.2		
					289.7	33.8	21.5	11.4			18.8	27.2

Abbreviations: Site ID: site identification; Strike/dip: strike azimuth and dip of bed; R/N: samples show reversed and normal polarity; N0: number of samples in the calculation; Dg, Ig (Ds, Is): declination and inclination of direction in *in situ* (after tilt-correction); Φ(G), λ(G) (Φ(S), λ(S)), longitude and latitude of VGP in geographical (stratigraphic) coordinates.

* The site-mean direction was rejected for further consideration either the larger uncertainty or deviated far from the group mean direction.

Carboniferous Chouniugou Formation (Shao and Qiu, 1988; Song and Li, 2009). Song and Li (2009) suggested that this mineralising event was related to hydrothermal activity and associated metamorphism, with hydrothermal temperatures that reached up to 250–300 °C. This indicates that the remagnetisation of the early Carboniferous Chouniugou Formation was caused by hydrothermal metamorphism. In addition, early Carboniferous samples from the study area have magnetic fabrics that indicate that the minimum susceptibility axis is not perpendicular to bedding, suggesting that the primary magnetic fabric was overprinted by remagnetisation (Fig. 8). We therefore suggest that hydrothermal fluids caused the remagnetisation of the early Carboniferous limestone, probably during the late Carboniferous and early Permian.

4.2. Late Paleozoic paleomagnetic poles of the Alashan Terrane

Recent research has identified Permian to Triassic paleomagnetic poles for the ALT (Zhu and He, 1987; He et al., 1988; Meng, 1992; Wu et al., 1993; Huang et al., 2001; Li et al., 2004; Liu et al., 2010; Table 5). These data can be divided into two groups using paleomagnetic reliability criterion, with one group from a very narrow area around Yumen, in the northwest of the Hexi Corridor; this group includes late Permian poles (Meng et al., 1990, Meng, 1992; Li et al., 2004) and an Early Triassic pole (Li et al., 2004; Table 5) for this area. Another group of late Permian (this study) and Early–Middle Triassic paleomagnetic poles (Liu et al., 2010) for the southeastern part of the ALT has a systematic

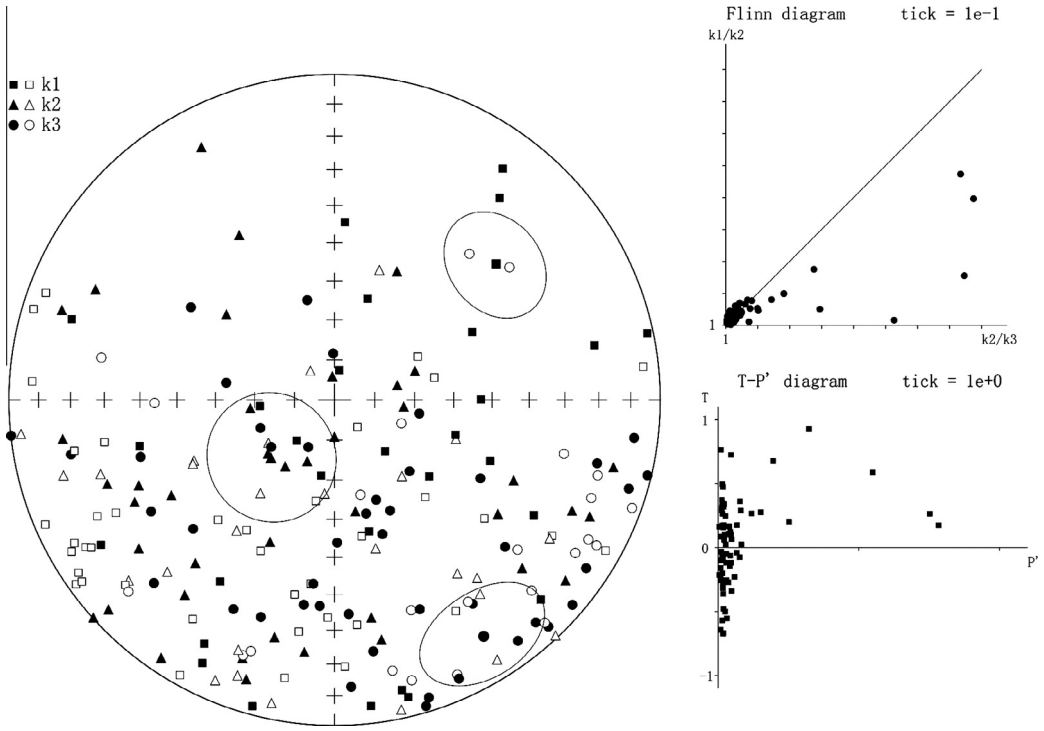


Fig. 8. Stereographic projection of AMS data for early Carboniferous limestone samples from the Xiaoweichuan section shown with stratigraphic coordinates; k_1 = maximum, k_2 = intermediate, k_3 = minimum, k_2/k_3(F) = magnetic foliation, k_1/k_2(L) = magnetic lineation, P' = anisotropy degree, T = shape parameter, $k_1 = -0.5711$, $k_2 = -1.0136$, $k_3 = -1.4153$, $p' = 2.504$, $T = 0.264$.

deflexion with respect to those from the Yumen area. More recent research by Sun et al. (2012) identified several paleomagnetic results from the Yumen area that included an Early Cretaceous pole for the Hanxia section (39.77°N, 97.17°E) that was obtained very close to the sampled Permo-Triassic section at Yumen, but records a clockwise rotation compared with the coeval pole from Eurasia and/or the NCB (Table 5). In comparison, Pliocene paleomagnetic data from another close section at Hongliuxia (39.96°N, 97.1°E) are indicative of only a minor rotation. The different declination obtained from these two results ($D_{Cretaceous} = 26.1°$ and $D_{Pliocene} = 355.4°$) also indicates that a post-Early Cretaceous clockwise rotation in the Yumen area. We therefore consider that the Permo-Triassic poles from the Yumen area most probably underwent a clockwise rotation of >25° related to movement on the Altyn Tagh Fault. This also suggests that the Permo-Triassic poles from the Yumen area underwent a local rotation relative to the eastern part of the ALT. In addition, the early Permian pole obtained from the Zhongwei, Shandan, and Hangwula areas in the western part of the ALT is consistent with the poles from the eastern part of the ALT (Wu et al., 1993; Table 5).

Table 5 lists late Permian and Triassic poles from the NCB (Yang et al., 1998). If recent paleomagnetic research on Late Devonian sediments in central-eastern Inner Mongolia is taken as correct, then the Paleo-Asian Ocean closed before the Late Devonian (Zhao et al., 2013), indicating that the late Carboniferous Inner Mongolian pole can be used for the NCB (Zhao et al. 1990). Fig. 9 indicates that the poles from the ALT are distinguished with the coeval NCB poles within a 95% confidence level. However, if the Carboniferous, late Permian (this study), and Early–Middle Triassic (Liu et al., 2010) poles of the eastern ALT are rotated counterclockwise by 32° around a Euler pole at 44°N, 84°E, then the coeval poles from the ALT and NCB consistently overlap (Fig. 9).

These data suggest that the Hexi Corridor–Alashan Terrane converged with the NCB by a counterclockwise rotation of 32° after the Middle Triassic. The fact that the paleomagnetic poles from the ALT and the NCB are compatible from the Early Cretaceous onwards (Sun et al., 2001) suggests that the ALT and NCB amalgamated before the Early Cretaceous.

4.3. Late Paleozoic and early Mesozoic Tectonic evolution of Northern China

The ALT is considered a key area that connects the Mongolia–NCB, Tarim, and Qaidam blocks with the Qilian Orogenic Belt of northern China. A clear difference in late Permian paleo-latitudes between the ALT–NCB and Tarim blocks suggests that the Tarim Block collided with Kazakhstan and Siberia by the end-Permian, whereas the NCB and the Mongolian Orogenic Belt were distant from Siberia during the Permian–Triassic (Enkin et al., 1992; Yang and Besse, 2001). This led to Enkin et al. (1992) suggesting that a megashear could reconcile these differences in paleo-latitude. The pre-Miocene tectonics of this area may be obscured by the widespread deposition of Quaternary sediments on top of the ALT, the development of a giant Jurassic–Cretaceous thrust–nappe structure system that overthrust Carboniferous units (in the allochthonous nappe) onto Cambrian units in the western margin of the Ordos Basin (Helanshan–Zhouzishan Mountains; Zhang et al., 2008), and a large crustal shortening event that affected the West Qilian Ranges and the Qaidam Basin that initiated during the late Miocene (Meyer et al., 1998). Thus, the thrust–nappe system may have destroyed the evidence of any early shearing between the western Ordos Basin and the ALT, although a large-scale ductile deformation zone was recently identified in the NE–SW-trending Bayanwula Shan–Lang Shan area; this zone is thought to represent a tectonic boundary between the NCB and the ALT (Zhang et al., 2013).

Combining the late Permian (260 Ma) paleomagnetic data for eastern Asia with our current knowledge of orogenic processes has enabled us to create a reconstruction that enables a better understanding of Asian tectonics (Fig. 10). We use the late Permian data of Yang and Besse (2001) for the north and south China blocks,

Table 5
Carboniferous and Mesozoic paleomagnetic data from the Hexi Corridor-Alashan Terrane and NCB.

Age	Sample site	Samples	Paleopole (°)		A95	Plat (°)	Test	Reference
			Lat.	Long.				
Hexi Corridor and Alashan Terrane								
N2	Hongliuxia (40.0°N,97.1°E)	21	78.7°N	298.0°E	8.4	29.3°N	–	Sun et al. (2012)
K1	Hanxia (39.8°N,97.2°E)	91	66.7°N	203.3°E	7.9	30.3°N	–	Sun et al. (2012)
T_{1-2}	Jingtai (37.2°N,104.3°E)	226	39.6°N	12.7°E	5.2	21.6°N	F,R	Liu et al. (2010)
T_1	Yumen (40.3°N, 97.0°E)	26	63.7°N	351.1°E	12.3°	29.1°N	R	Li et al. (2004)
P_2	Yumen (40.3°N, 97.0°E)	33	54.7°N	359.9°E	12.2°	28.2°N	R	Li et al. (2004)
P_2	Yuman (39.6°N, 98°E),Sunan (39°N, 99.6°E)	56	41.4°N	358.6°E	2.7°	17.5°N	F,R,C	Meng (1992)
P_2	Shizuishan (39.2°N,106.3°E)	68	27.2°N	18.8°E	12.0°	18.6°N	F	This Study
P_1	Zhongwei, Shandan, Hangwula	22	23.9°N	21.2°E	12.8°	21.9°N	F	Wu et al. (1993)
P_1	Yumen (40.3°N, 97.0°E)	8	30.5°N	18.6°E	23.3°	27.6°N	R	Li et al. (2004)
P_1	Alashan (41.7°N, 102.6°E)	28	64.3°N	246.8°E	9.5°	19.7°N	–	Zhu and He (1987)
P	Jingtai-Gulang (37.2°N, 103.5°E)	13	55.6°N	233.7°E	5.4°	12.0°N	F,R	He et al. (1988)
C_{1-2}	Zhongwei (37.2°N, 105.4°E)	39	10.5°N	14.0°E	6.2°	6.0°N	F	Huang et al. (2001)
C_3-P_1	Zhongwei (37°N, 105°E)	48	13.1°N	11.0°E	7.0°	4.5°N	F	This Study
North China Block (reference point: 38°N and 112°E)								
Cenozoic		5*	88.7°N	202.9°E	4.7	38°N		Yang et al. (1998)
K		3*	80.7°N	204.9°E	9.5	36°N		Yang et al. (1998)
J_3		2*	69.4°N	221.8°E	15.4	28°N		Yang et al. (1998)
J_2		5*	76.8°N	229.7°E	5.6	31°N		Yang et al. (1998)
J_1		1*	82.4°N	286.0°E	6.8	30.4°N		Yang et al. (1998)
T_3		1*	62.3°N	7.7°E	3.8	27.0°N		Yang et al. (1998)
T_3		1*	70.1°N	9.4°E	5.8	31.4°N		Yang et al. (1998)
T_2		7*	61.0°N	2.0°E	2.9	24.1°N		Yang et al. (1998)
T_1		6*	56.9°N	353.7°E	4.6	18.2°N		Yang et al. (1998)
P_2		11*	50.3°N	355.1°E	5.7	14.2°N		Yang et al. (1998)
P_2		(11 sites)	51.9°N	341.7°E	6.8	9.8°N		Zhao et al. (1990)
C_3		29 samples	32.6°N	338.1°E	4.4	7.4°S		Zhao et al. (1990)

Notes: A95: precision parameters and half-angle of cone of 95% confidence about the mean palaeopole; lat./long.: latitude/longitude of the palaeopole; Plat: palaeolatitude; F, R and C: fold test, reversal test and regional coherent test;
* Denotes the number of paleopoles used.

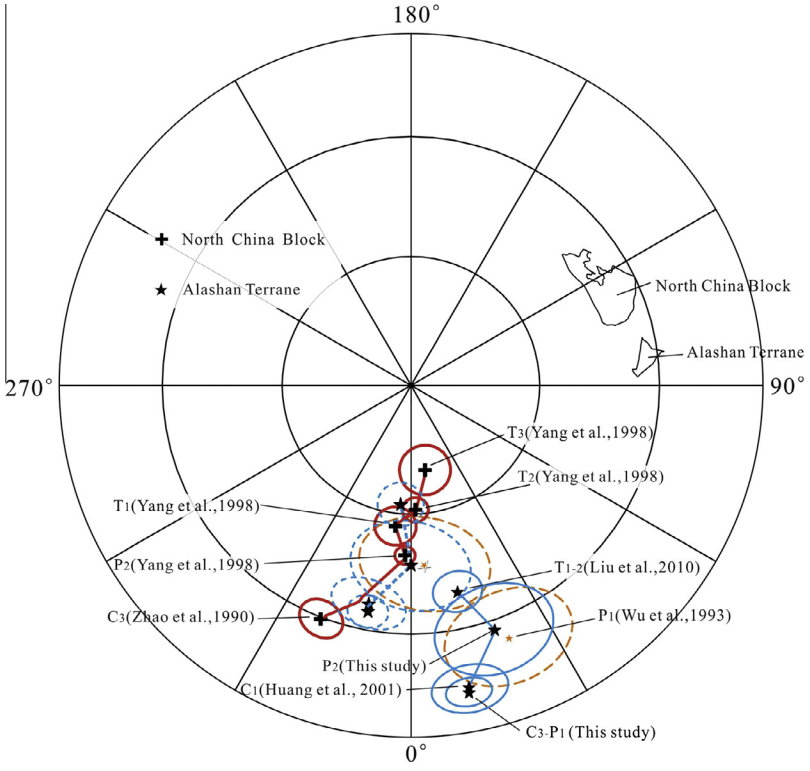


Fig. 9. Equal-area projection of early Carboniferous, late Carboniferous–Permian, late Permian, and Triassic apparent polar wander paths for the ALT (indicated by a pentagon) and the NCB (indicated by a cross); dotted-line poles represent eastern ALT paleopoles (pentagon) after 32° counterclockwise rotation around a Euler pole at 44°N, 84°E. Note that the ALT was rotated to the NCB in present coordinate about the Euler pole at 44°N, 84°E, CW rotation 32°.



Fig. 10. Paleomagnetic-based paleogeographic reconstruction of major Asian blocks in the late Permian; TAR = Tarim Block, Q = Qaidam Block, ALT = Alashan Terrane, NCB = North China Block, SCB = South China Block, MONG = Mongolian fold belt, INC = Indochina Block, LA = Lhasa Block, Qi = Qiangtang Block. Red lines indicate presumed subduction zones and strike-slip faults. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the data presented by Li et al. (1988) for the Tarim Block, data from Xu et al. (2011) for the Qaidam Block, data from Huang et al. (1992) for the Qiangtang Block, and the data of McElhinny et al. (2003) for Gondwana. Recently obtained late Permian paleomagnetic data from north Kazakhstan indicate that no substantial motion occurred between the north Kazakhstan and the Baltica areas that are separated by the Urals (Bazhenov et al., 2008), meaning that the Kazakhstan Block moved rigidly with Siberia; as such, we use the Siberian–Kazakhstan reference pole of Bazhenov et al. (2008).

This reconstruction also assumes that amalgamation of the NCB and the Mongolian Orogenic Belt was almost complete by the late Permian in the area from Solonker eastwards to the Xar-Moron River in southeastern Inner Mongolia, and in the Yanji area of central Jilin Province (Zhu et al., 1998; Li, 2006), with Cenozoic crustal shortening in the Qilian Shan Orogenic Belt, and about 700 km of Cenozoic left-lateral strike slip faulting along the Red River fault zone (Yang and Besse, 1993). This reconstruction is fairly similar to the reconstruction presented by Enkin et al. (1992), although we separate the ALT from the NCB, and we modified the orientation of the ALT by fitting Carboniferous, Permian and Early–Middle Triassic poles through a counterclockwise rotation of 32° around an Euler pole at 44°N, 84°E with the coeval poles of the NCB.

The Altaids is one of the largest known accretionary orogenic collages and is associated with considerable Phanerozoic juvenile crustal growth (Sengör and Burtman, 1993; Jahn et al., 2000; Jahn, 2004; Xiao et al., 2009b). This collage was constructed by convergence between the southern active margin of the Siberian Craton to the north and the northern margins of the Tarim and North China cratons to the south (Xiao et al., 2009b). The ALT is located within the central part of the southern Altaids, and two ophiolitic melange belts, the Engger Us and Qagan Chulu ophiolite belts, have been identified in the northern ALT; these belts provide geological information on the tectonic evolution of this area (Wang et al., 1992, 1993; Wu and He, 1992; Wang et al., 1995; Zheng et al., 2014). The Engger Us ophiolite belt represents a major Paleo-Asian Ocean suture within the northern ALT and the Qagan Chulu ophiolite belt represents a back-arc basin (Zheng et al., 2014). Geochronological analysis indicates that final closure of Paleo-Asian Ocean may have occurred after the early Permian (Zheng et al., 2014), and most probably between the end Permian and the Middle Triassic. These results are supported by 250–213 Ma collision-related alkali-rich intrusives located within the northern ALT and the Beishan Belt (Ren et al., 2005a,b). The geology of this area also suggests that collision between microcontinents of the southern Altaids and the ALT did not finish until the Late Triassic (Feng et al., 2013). However, Mesozoic paleomagnetic poles for the joint North China–Mongolia Block and Siberia are distinct from each other, indicating that this collision may have initiated in the west, and was followed by progressive rotation and suturing in the east until the Late Jurassic (Zhao et al., 1990; Enkin et al., 1992).

The southeastern part of the ALT is connected to the Qinling Orogen, an area of multi-stage tectonism since the middle Paleozoic, and final suturing between the NCB and the South China block (SCB) was accomplished by a large Late Triassic to Middle Jurassic counterclockwise rotation of the NCB by 47° ± 8° (Yang et al., 1992). The Mianlue suture along the southern margin of the Qinling Orogenic Belt provides evidence of the processes involved in collision between the NCB and the SCB (Dong et al., 2012). The geology of this suture indicates that the Mianlue ocean basin began to open in the Middle to Late Devonian, and expanded between the

Carboniferous and Permian (Zhang et al., 2001; Sun et al., 2002; Lai et al., 2004; Wu and Zheng, 2012; Song et al., 2013), before the onset of northward subduction of this oceanic crust, as evidenced by island-arc volcanism and subduction-related deformation between 251 and 217 Ma (Dong et al., 2011). A series of syn-collisional 207–201 Ma monzogranites and biotite granites constrains the timing of this event, and 189 Ma granodiorites provide evidence of the timing of post-orogenic collapse of the Qinling Orogenic Belt (Dong et al., 2011, 2012; Wang et al., 2013). Geochronological and geochemical data from the Kokoxili and Yushu regions of central and eastern Tibet provide evidence of coeval magmatism in both the Jinsha and Kunlun sutures, suggesting that two subduction zones were simultaneously active during the Late Triassic (Roger et al., 2003).

These results suggest that the ALT may have amalgamated with the NCB through a series of Late Triassic and/or Early Jurassic counterclockwise rotations and transform faulting. Ductile shearing along the boundary between the ALT and the NCB is evident along the NE–SW-trending Bayanwula Shan–Lang Shan area, although this deformation event has been dated to ca. 250 Ma (biotite and muscovite $^{40}Ar/^{39}Ar$ ages; Zhang et al., 2013), earlier than suggested by the data presented here. This amalgamation may be related to north-dipping oblique subduction of the SCB and northward subduction of the Kunlun Terrane. This tectonism is consistent with the pre-Early Jurassic major deformation and décollement tectonics recorded in the Songpan–Garze Fold Belt (Roger et al., 2004).

5. Conclusions

This study reports new paleomagnetic data for early Carboniferous and late Permian sediments within the eastern ALT. Systematic thermal demagnetisation yielded a site-mean direction of the middle temperature component of the early Carboniferous sediments of Ds = 283.0°, Is = 9.0°, ks = 47, and α_{95} = 8.9°, with a corresponding pole at λ = 13.1°N, Φ = 11.0°E (A_{95} = 7.0°). These data may represent a remagnetisation event associated with interaction between these sediments and late Paleozoic orogenic fluids within the Qilian Orogenic Belt. The late Permian sediments have a site-mean direction of Ds = 289.7°, Is = 33.8°, ks = 21.5, and α_{95} = 11.4° after tilt correction, with a corresponding pole at λ = 27.2°N and Φ = 18.8°E (A_{95} = 12.0°), with a paleolatitude of 18.6°N ± 12.0°. These data, combined with previous paleomagnetic data for the ALT, yield a new APWP that includes early Carboniferous, late Carboniferous–early Permian, late Permian, and Early–Middle Triassic poles. Comparison with coeval poles from the NCB indicates rotation of the poles of the eastern ALT counterclockwise by 32° around an Euler pole at 44°N, 84°E, yielding consistent overlapping of coeval poles from the ALT and the NCB. These data indicate that the ALT may have amalgamated with the NCB through a series of counterclockwise rotations and transform faulting during the Late Triassic and Early Jurassic, the first step in creating eastern Asia.

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